

Malicious IPs Analysis Report 2026

VirusTotal Multi-Engine Platform · 200 IP Address Records

Submitted by

Izma Amin

Student ID: 70332

Course: Probability and Statistics

Analysis Date: June 9, 2026

§1 Abstract

This report presents an analysis of the Malicious IPs dataset comprising 200 IP address records collected via VirusTotal's multi-engine threat intelligence platform during 2026. The analysis focuses on predicting threat severity (Low, Medium, High) using geographic distribution, autonomous system (ASN) classification, Tor network membership, and threat voting metrics.

The Random Forest Regressor demonstrated superior performance with $R^2 = 0.844$ and $RMSE = 0.132$ with excellent cross-validation stability (5-fold CV $R^2 = 0.842$, $\sigma = 0.017$). Malicious_Votes dominates threat severity prediction at an extraordinary 86.76% importance. European infrastructure accounts for 62.0% of malicious IPs, with East European countries (Russia, Romania, Ukraine, Bulgaria) accounting for 48.5% of total infrastructure.

§2 Introduction

Malicious IP infrastructure serves as source addresses for command-and-control (C2) communications, malware distribution, brute-force attack origination, botnet coordination, and data exfiltration endpoints. Unlike domain infrastructure requiring DNS resolution, IP addresses provide direct network accessibility and represent geolocation-specific threat signals.

§3 Dataset Description

3.1 Geographic Distribution — Continental

Continent	Count	Percentage	Primary Threats
Europe (EU)	124	62.0%	DOMINANT — East European APT Infrastructure
Unknown	48	24.0%	VPN/Proxy masked origins
Asia (AS)	24	12.0%	Secondary — Chinese APT, Botnet Hosting
Africa (AF)	4	2.0%	Minimal representation
TOTAL	200	100.0%	

3.2 Top Countries (where identified)

Rank	Country	Code	IP Count	Infrastructure Type
1	Russia	RU	34	State-sponsored, Bulletproof Hosters
2	Romania	RO	28	Bulletproof hosting, carding infrastructure
3	Ukraine	UA	19	Hosting providers, hacking-as-a-service

4	Bulgaria	BG	16	IPTV piracy, phishing infrastructure
—	Unknown	UN	48	VPN/Proxy masking

East European Concentration: Top 4 countries (RU, RO, UA, BG) account for 97 IPs (48.5% of dataset), confirming dominance of post-Soviet/Eastern European threat actor infrastructure.

3.3 Autonomous System (ASN) Distribution — Top 10

ASN	Organization	IP Count	%	Infrastructure Type
AS3269	Telecom Italia	18	9.0%	Compromised legitimate ISP
AS8452	Telenor ASA	15	7.5%	Compromised legitimate ISP
AS12389	Rostelecom	12	6.0%	State-associated Russian ISP
AS16509	Amazon Web Services	11	5.5%	Compromised cloud accounts
AS3352	Telenor Business	9	4.5%	Business ISP compromised
AS8359	MTS Telecom	7	3.5%	Russian national telecom
AS1299	Telia Company	6	3.0%	Scandinavian ISP compromise
AS6830	Liberty Global	5	2.5%	UK ISP compromised instances
AS20040	China Unicom	4	2.0%	Chinese national ISP
AS3561	Savvis/CenturyLink	4	2.0%	US ISP compromised instances

3.4 Tor Network Membership

Tor Status	Count	Percentage	Threat Interpretation
Non-Tor (Clearnet)	189	94.5%	Clearnet malicious infrastructure
Tor Exit Node	11	5.5%	Tor network anonymity infrastructure

3.5 Threat Severity Distribution

Severity	Count	Percentage	Description
Low	174	87.0%	Minimal threat signals; limited engine verdicts
Medium	14	7.0%	Moderate threat signals; mixed verdicts
High	12	6.0%	Strong threat signals; high confidence malicious
TOTAL	200	100.0%	

3.6 Voting Metrics Distribution

Metric	Mean	Std Dev	Min	Max
Malicious_Votes	1.50	3.21	0	16

Harmless_Votes	43.2	21.8	1	84
Suspicious_Votes	0.21	0.89	0	5
Undetected_Votes	46.0	22.9	4	90
Total_Reports	8.34	14.2	0	68
Reputation_Score	-2.02	18.6	-80	40

§4 Correlation Analysis

Feature Pair	Correlation	Interpretation
Malicious_Votes ↔ Threat_Severity	+0.897	VERY STRONG positive correlation
Reputation_Score ↔ Threat_Severity	-0.671	Strong negative correlation
Continent (Europe) ↔ Threat_Severity	+0.156	WEAK positive correlation
TOR_Node ↔ Threat_Severity	+0.089	NEGLIGIBLE correlation

§5 Regression Models — Five Models Evaluated

Model	RMSE	R ² Score	Overfitting	Utility
Random Forest (BEST)	0.1320	0.8440	Low	YES — Recommended
Gradient Boosting	0.0001	1.0000	EXTREME	No — Overfitting
Lasso (α=0.01)	0.2890	0.3190	Low	Limited
Ridge (α=1.0)	0.3010	0.2410	Low	Limited
Linear Regression	0.3210	0.1680	Low	Limited

§6 Model Evaluation

6.1 Random Forest Performance (Recommended)

- Training R²: 0.849 | Test R²: 0.844 | Train-Test Gap: 0.5% (excellent)
- RMSE: 0.132 | Mean Residual: +0.0067 (slightly optimistic bias)
- 5-Fold CV Mean R²: 0.842 (σ = 0.017) — excellent stability
- Cross-Validation Fold Results: 0.839, 0.845, 0.841, 0.843, 0.846

6.2 Feature Importance (Random Forest) — Extraordinary Malicious_Votes Dominance

Rank	Feature	Importance	Percentage	Interpretation
1	Malicious_Votes	0.8676	86.76%	OVERWHELMINGLY DOMINANT

2	Undetected_Votes	0.0540	5.40%	Secondary voting signal
3	TOR_Node	0.0310	3.10%	Tor network membership
4	Reputation_Score	0.0200	2.00%	Reputation metric limited
5	Is_Top_10_ASN	0.0130	1.30%	Major ASN infrastructure
6	Continent	0.0085	0.85%	Geographic location weak signal
7	Total_Reports	0.0031	0.31%	Negligible
8	Detection_Confidence	0.0018	0.18%	Minimal

§7 Discussion & Conclusion

Malicious_Votes' 86.76% importance is remarkably concentrated, indicating that threat voting directly determines severity classification with minimal contribution from geographic, network, or structural features. IP reputation consensus is more concentrated than domain reputation (86.76% vs 48.3%).

7.1 Geographic Features Provide Minimal Predictive Signal

Despite the European infrastructure concentration (62%), continental geolocation contributes only 0.85% to threat severity predictions. Sophisticated threat actors deliberately distribute infrastructure across multiple jurisdictions to avoid geographic fingerprinting.

7.2 Strategic Recommendations

- Deploy Random Forest Regressor ($R^2=0.844$) for operational IP triage
- Prioritize VirusTotal Malicious_Votes as primary threat severity indicator
- Implement SMOTE oversampling for improved high-confidence threat detection
- Integrate model predictions into SOC alerting and network blocking workflows
- De-prioritize geographic and ASN-based filtering in favor of voting-based classification
- Incorporate temporal features: days since last malicious activity, BGP hijacking detection

Report Metadata: Dataset: 200 records · Best Model: Random Forest $R^2=0.844$ · Feature Leader: Malicious_Votes 86.76% ·
Date: June 9, 2026